The finite boundary for two equal blocks

Bounded proof component for the transitive rank theorem

12 October 2026

We establish the finite obstruction used in the two-block argument. The classification of the possible groups in this statement is a separate obligation: no conclusion about other groups is implicit in the calculation below. In particular, this component is not a proof of the full transitive rank theorem.

Let $W = S_k \wr C_2$ act on two blocks of size k, and let r be the permutation exchanging their corresponding letters. For a subgroup $H \leq S_k$ and a character $\chi : H \to C_2$, put

$$D_\chi(H) = \{(h, h)r^{\chi(h)} : h \in H\}.$$

The two symmetric factors in a row $S_a \times S_b$ of the table below act on disjoint sets of sizes a, b in one block. The two transitive degree-six actions are the literal permutation groups numbered 9 and 11 in the GAP transitive-group catalogue; their orders are 36 and 48, respectively. The action, rather than the abstract group alone, is part of the specification.

Lemma 1. *Let H be one of the ten actions in Table 1, of degree k. Take an involutory string C-marking $(h_1, \ldots, h_{k-2})$ of H, and let C be its image in $D_\chi(H)$. Split this ordered list as C_L, C_R, requiring the two lists to commute elementwise. There is no tuple*

$$(C_L, x, r, y, C_R)$$

of independent involutions that generates a transitive subgroup of W, satisfies the string commutation relations, has both x and y noncommuting with r, and has r at least as close to an endpoint as every other generator exchanging the blocks.

Proof. We give the complete finite enumeration and its exhaustion argument. The accompanying program uses exact finite permutation groups.

First enumerate the involutory string C-markings of H. At a prefix $(h_1, \ldots, h_j)$, an eligible next generator is an involution of H outside the prefix subgroup and commuting with $h_1, \ldots, h_{j-1}$. We discard a prefix if one of its generators belongs to the subgroup generated by the others. Every subfamily of an independent family is independent, so this discards no desired tuple. Also, if the current subgroup has index less than 2^{k-2-j} in H, the prefix cannot acquire the required number of independent generators: each strict subgroup enlargement has index at least two.

Simultaneous conjugation by $N_{S_k}(H)$ preserves every condition. At each step choose one orbit representative under the pointwise stabilizer of the prefix under conjugation, then replace that stabilizer by its centralizer of the chosen generator. Induction on the prefix length shows that every ordered marking is represented. At length $k - 2$, require generation of H and test the intersection property for every pair of subsets of the marking. The test forms the actual generated subgroups and their intersections, without a quotient or an assumed presentation.

For each surviving marking enumerate all 2^{k-2} assignments of its generators to C_2. Retain exactly those assignments for which `GroupHomomorphismByImages` constructs a homomorphism. Since the marking generates H, these are exactly its characters. Construct the corresponding diagonal lifts explicitly as permutations of $2k$ letters. Try every cut between the lifts, including

the two empty-end cuts, and require elementwise commutation across it. Reject a cut and character only if an already fixed block-exchanging generator is closer to an endpoint than r.

The string relations require x to centralize all of C_R and all but the last member of C_L. Similarly, y must centralize all of C_L and all but the first member of C_R. Compute these two centralizers inside W, enumerate their involutions, and impose noncommutation with r and the same endpoint restriction. We may choose one representative for x under the simultaneous centralizer of C and r in W: conjugation by this group preserves the complete candidate set for y. For each remaining pair require $xy = yx$ and transitivity. Finally test independence by deleting each generator in turn and comparing the resulting subgroup with the full group.

The exhaustive counts are in Table 1. No transitive completion occurs in the first eight cases. In the last two cases, all transitive completions fail independence. Thus no test of the full intersection property is needed to reject a completion, proving the lemma. □

Action of H	k	Markings	Transitive completions	Independent
$S_2 \times S_3$	5	4	0	0
$S_2 \times S_4$	6	12	0	0
$S_3 \times S_3$	6	1	0	0
$S_2 \times S_5$	7	4	0	0
$S_3 \times S_4$	7	6	0	0
$S_2 \times S_6$	8	8	0	0
$S_3 \times S_5$	8	2	0	0
$S_4 \times S_4$	8	9	0	0
TransitiveGroup$(6, 9)$	6	1	2	0
TransitiveGroup$(6, 11)$	6	12	152	0

Table 1: Complete bounded search. Markings are represented up to the normalizer conjugations described in the proof. Completion counts use the stated pruning of x; they are not counts of isomorphism classes.

The implementation is in `search.g`; `run.g` executes all ten cases, and `check.py` requires their exact summaries and successful completion. To check that the procedure can retain valid completions, it is also run with $k = 3$ and the empty marking of the trivial group. It recovers six independent transitive rank-three string C-markings after the same pruning. For example,

$$((5\ 6),\ (1\ 4)(2\ 5)(3\ 6),\ (1\ 2))$$

passes the full intersection test. A redundant triple of distinct transpositions in S_3 supplies negative controls for independence and the intersection property.

The proof depends on GAP's exact finite permutation-group algorithms and its degree-six transitive catalogue. It does not supply a Lean formalization. The separate reduction of an arbitrary two-block group to these ten actions, the endpoint cases, and all other branches of the transitive rank theorem remain outside this lemma. Normalizing the commuting generators by powers of the block swap already appears in Cameron, Fernandes, Leemans and Mixer, *String C-groups as transitive subgroups of* S_n, arXiv:1410.5863v1 (2014), proof of Proposition 2.1, Case 3. We retain that step. The present contribution is the explicit character description and exhaustive finite analysis with two noncommuting neighbors, within the common string-separator argument. It neither claims a new open-conjecture solution nor cites the target rank theorem as a premise.